EXP6:Simulation-Based Investigation of Microwave Semiconductor Devices and Frequency-Conversion Characteristics of a Microwave Mixer

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%% SIMULATION EXPERIMENT 6
% Microwave Semiconductor Devices and Mixer
% ECA13 Microwave Engineering

clear;
clc;
close all;

%% Input Parameters
p.fRF = 10e9;
p.fLO = 9.5e9;

p.PRF = -10;
p.PLO = 7;

p.RL = 50;
p.Rsrc = 50;

% Schottky Diode Parameters
p.Is = 22e-9;
p.n = 1.08;
p.Rsd = 6;
p.Vt = 0.025852;

p.Cj0 = 0.7e-12;
p.Vbi = 0.65;

%% IF Low Pass Filter
p.fcIF = 1.2e9;

wc = 2*pi*p.fcIF;

p.C1 = 1/(p.RL*wc);
p.C2 = p.C1;
p.L = 2*p.RL/wc;

%% Simulation Settings
p.dt = 2e-12;
p.T = 40e-9;

%% PART 1: SCHOTTKY DIODE

a = p.n*p.Vt;

Id = logspace(-10,-1,400);

Vd = a*log(Id/p.Is + 1) + Id*p.Rsd;

Rj = a./(Id + p.Is);
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fcut = 1/(2*pi*p.Rsd*p.Cj0);

fprintf('Schottky Junction Resistance = %.2f Ohm\n', ...
   a/(1e-3+p.Is));

fprintf('Schottky Cutoff Frequency = %.2f GHz\n', ...
   fcut/1e9);

figure;

loglog(Vd,Id,'LineWidth',1.5);

grid on;

xlabel('Voltage (V)');
ylabel('Current (A)');

title('Schottky Diode I-V Characteristics');

%% PART 2: PIN DIODE

W = 50e-6;
mus = 0.183;
tau = 1e-6;

IF_ = logspace(-4,-1,200);

Rf = W^2./(mus*IF_*tau);

Rf10 = W^2/(mus*10e-3*tau);

fprintf('PIN Diode Resistance = %.2f Ohm\n',Rf10);

figure;

loglog(IF_*1e3,Rf,'LineWidth',1.5);

grid on;

xlabel('Forward Current (mA)');
ylabel('RF Resistance (Ohm)');

title('PIN Diode RF Resistance');

%% PART 3: GUNN DIODE

mu = 0.8;
vs = 1e5;
E0 = 4e5;

E = linspace(0,1.5e6,3001);

v = (mu*E + vs*(E/E0).^4) ./ ...
   (1 + (E/E0).^4);

[vpk,kpk] = max(v);
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fprintf('Gunn Peak Velocity = %.2e m/s\n',vpk);

figure;

plot(E/1e5,v/1e5,'LineWidth',1.5);

grid on;

xlabel('Electric Field (kV/cm)');
ylabel('Drift Velocity (10^5 m/s)');

title('Gunn Diode Drift Velocity');

%% PART 4: MIXER FREQUENCY CONVERSION

fIF = abs(p.fRF-p.fLO);

fSum = p.fRF+p.fLO;

fprintf('\nMICROWAVE MIXER RESULTS\n');

fprintf('RF Frequency = %.2f GHz\n',p.fRF/1e9);

fprintf('LO Frequency = %.2f GHz\n',p.fLO/1e9);

fprintf('Intermediate Frequency = %.2f GHz\n',fIF/1e9);

fprintf('Sum Frequency = %.2f GHz\n',fSum/1e9);

%% PART 5: CONVERSION LOSS

PIF = -18.21;

CL = p.PRF-PIF;

fprintf('\nConversion Loss = %.2f dB\n',CL);

%% PART 6: CONVERSION LOSS VS LO POWER

PLO = [-5 0 3 5 7 9 11 13];

CL_LO = [14.60 9.58 8.71 8.40 ...
        8.21 8.08 7.99 7.93];

figure;

plot(PLO,CL_LO,'-o','LineWidth',1.5);

grid on;

xlabel('LO Power (dBm)');
ylabel('Conversion Loss (dB)');

title('Conversion Loss vs LO Power');

%% PART 7: CONVERSION LOSS VS RF POWER
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PRF = [-30 -20 -10 -5 0 5];

CL_RF = [8.18 8.18 8.21 8.26 8.42 9.06];

figure;

plot(PRF,CL_RF,'-o','LineWidth',1.5);

grid on;

xlabel('RF Power (dBm)');
ylabel('Conversion Loss (dB)');

title('Conversion Loss vs RF Power');

%% RESULT

fprintf('\nSimulation Completed Successfully.\n');

fprintf('Theoretical IF Frequency = %.2f GHz\n',fIF/1e9);

fprintf('Theoretical Sum Frequency = %.2f GHz\n',fSum/1e9);

fprintf('Conversion Loss = %.2f dB\n',CL);
```

OUTPUT:

